

Srijan

Death Race

— 2026 —

Jadavpur University

Team **ApeX**

Our Team:

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Overview

This document outlines the technical specifications, design considerations, and cost analysis for our custom-built racing bot, designed to compete in the Death Race 2026. Embodying the “Madmax” wasteland theme, our bot is engineered for raw survival, combining salvaged ingenuity with high-torque electronics to withstand extreme track hazards, inclines, and aggressive competition.

1) Design Considerations

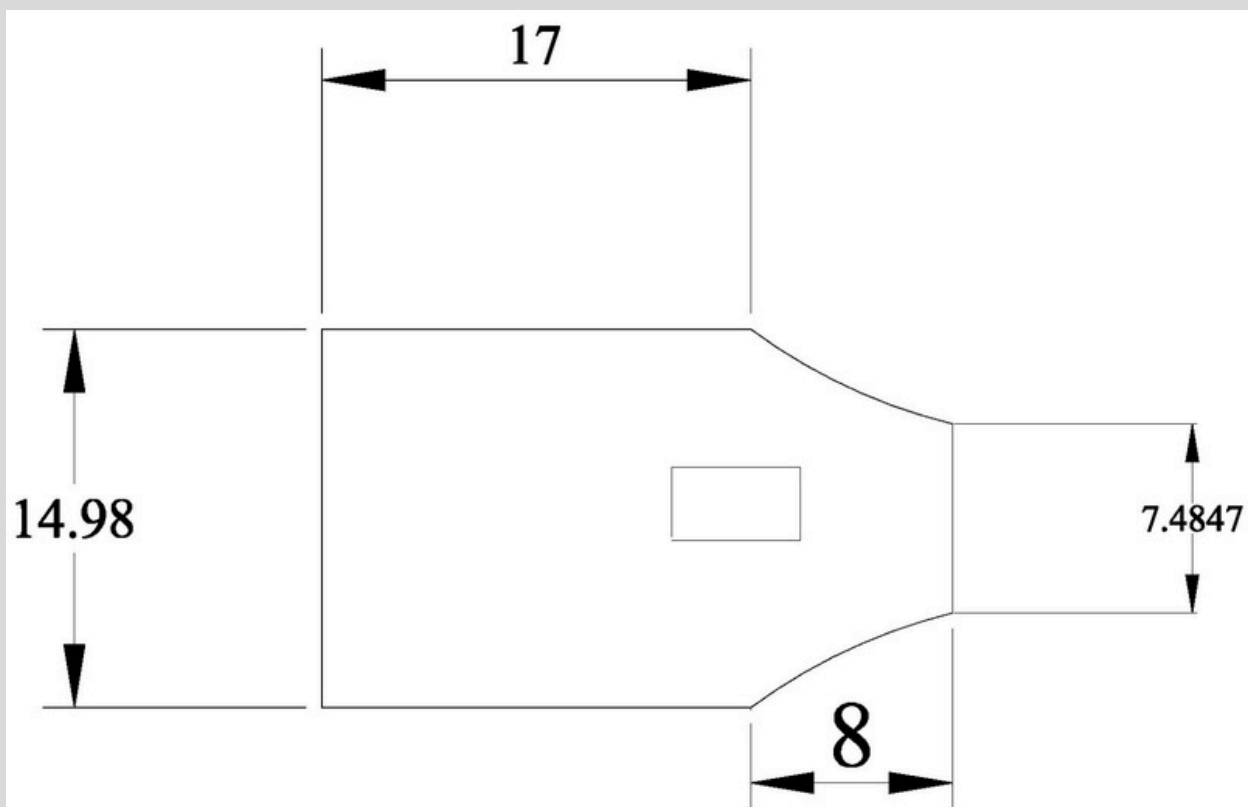
The bot is designed with a low center of gravity to prevent flipping on inclines and a robust custom steering mechanism to navigate sharp, demanding track obstacles.

Dimensions

In strict adherence to the competition limits (30 30 20 cm), our bot’s dimensions are designed to be:

- Length: 28 cm
- Width: 22.5 cm
- Height: 15 cm

CAD Models & Schematics

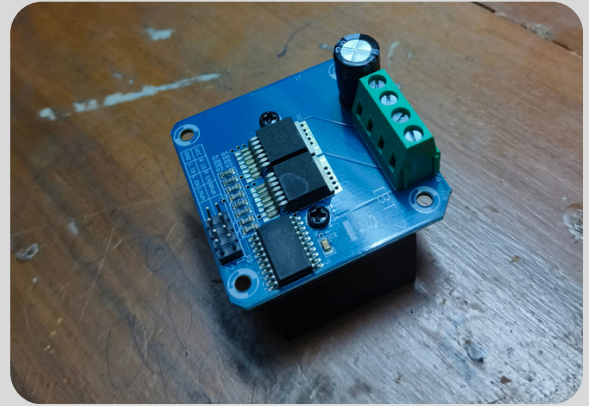


Chassis dimentions (all in cm)

2) Electronic Components

The drivetrain and control logic rely on a high-current, high-torque setup to push through the rigorous terrain.

- **Motors:** 2x 12V Johnson Geared DC Motors. Chosen for their immense stall torque, allowing the bot to push through obstacles and opposing bots effectively.



- **Motor Controller:** 1 BTS7960 43A High-Current Motor Driver Module. This H-bridge is crucial for handling the high stall currents drawn by the Johnson motors under heavy loads without burning out.

- **Battery:** 3S1P 2200mAh Li-Po pack (built using Pro-Range 21700-P42A cells). The 2200mAh capacity complies perfectly with the competition's maximum 5000mAh limit, while the 3S configuration (11.1V nominal) provides optimal voltage for the 12V motors.



3) Chassis Material and Justification

Material: Hard Plastic.

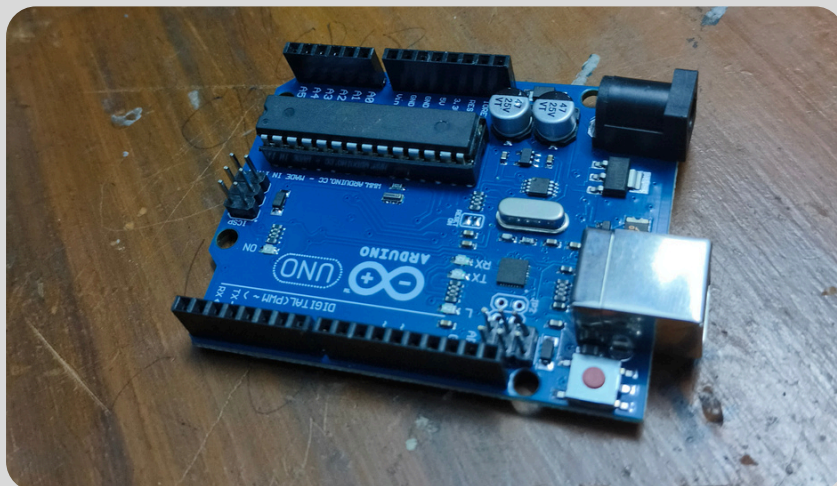
Justification: Hard plastic(exam board) was selected as the primary structural material due to its optimal balance of strength-to-weight ratio and cost. It is easy to cut, drill, and machine using standard hand tools. Furthermore, plastic acts as an excellent electrical insulator, mitigating the risk of short-circuiting the exposed Arduino or motor driver connections. It also provides enough flex to absorb minor impact shocks without permanently deforming.

4) Wireless Communication Method

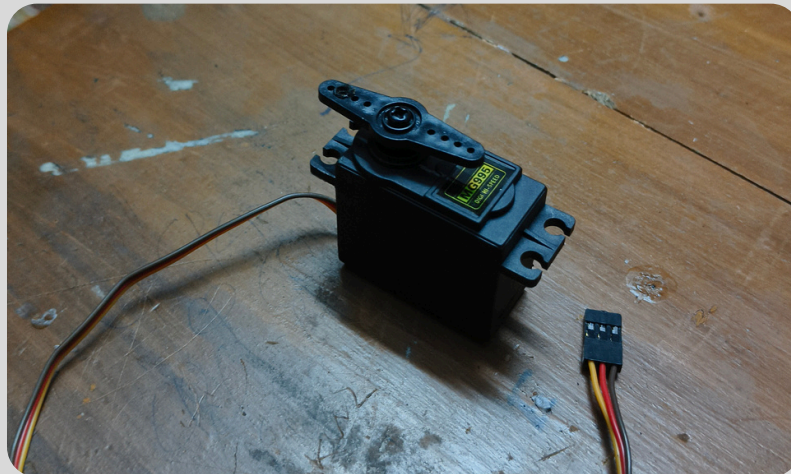
To comply with the "No Wired Connections" rule, control is maintained via a **Flysky FS-CT6B 6-Channel Transmitter paired with an FS-R6B Receiver**, operating on the reliable 2.4GHz AFHDS (Automatic Frequency Hopping Digital System) protocol. This setup provides a robust, interference-free wireless link with excellent range and minimal latency, which is critical for making precise, split-second steering and throttle adjustments amidst the RF noise of the competition arena.

5) Other Components Specifications

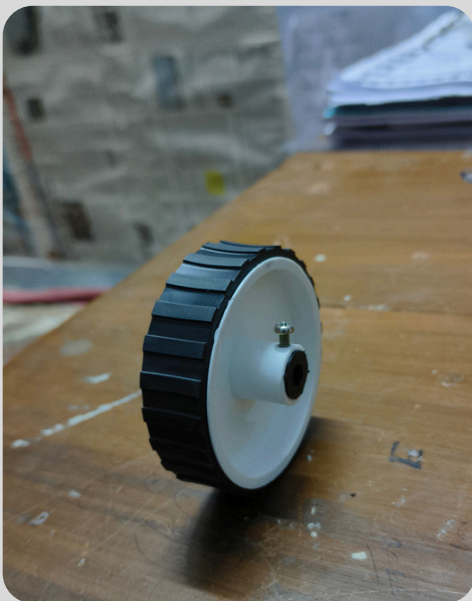
- **Microcontroller:** 1 Arduino Uno. Acts as the central brain, reading PWM signals from the FlySky receiver and mapping them to the BTS7960 motor driver and steering servo.



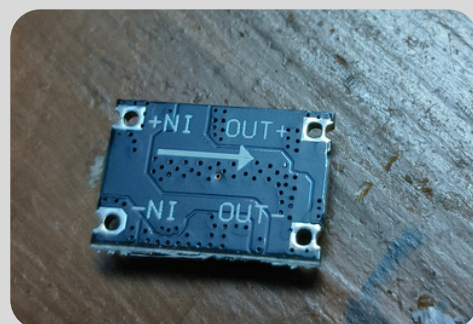
- **Steering Actuator:** 1 MG995R Metal Gear Servo. Provides the high torque required to turn the front wheels under load, with metal gears ensuring mechanical durability.



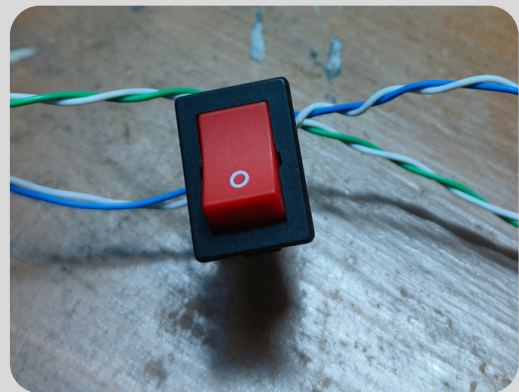
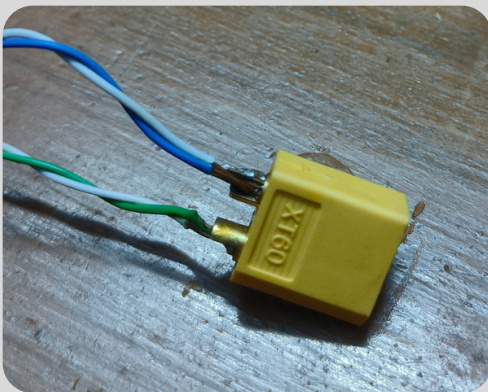
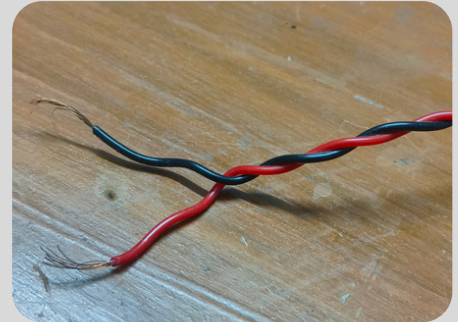
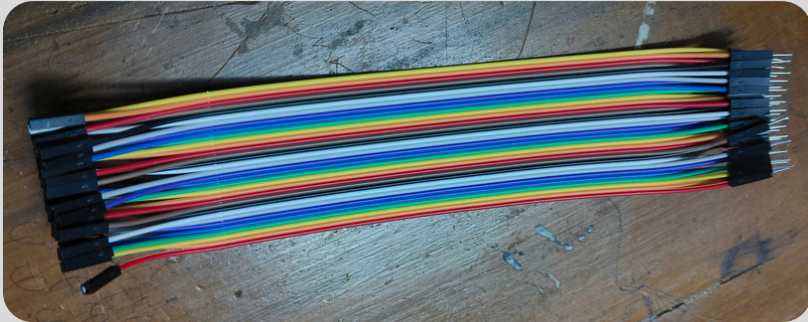
- **Wheels:** 4 Robotic Rubber Wheels (65–75 mm diameter) providing maximum grip and traction on dirt/track surfaces.



- **Power Regulation:** 1 UBEC 5V / 3A. Steps down the 11.1V Li-Po voltage to a safe, stable 5V to power the Arduino and the MG995R Servo without browning out the logic board during sudden current spikes.



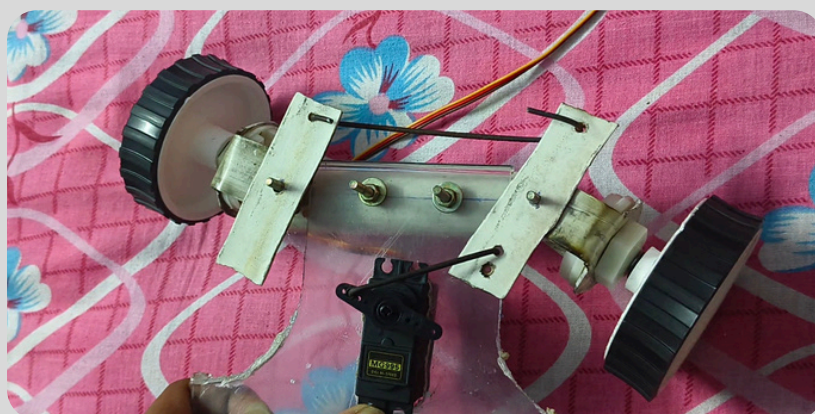
• **Wiring & Connectors:** Standard **multi-color ribbon jumper wires** are utilized for low-voltage signal routing between the Arduino, receiver, and motor driver. Embracing the upcycled “Madmax” theme, power distribution incorporates salvaged **twisted-pair cabling**. The main power line features a secure **XT60** connector for reliable battery attachment and an inline red rocker **switch** serving as an accessible master power cutoff.



6) Innovation Aspects

The most significant innovation in this build is the **custom-manufactured steering mechanism**. Instead of relying on expensive, off-the-shelf aluminum steering knuckles and tie rods, the steering linkage was engineered using repurposed materials:

1. PVC Linkages: Standard PVC pipes were heated, flattened, and drilled to create custom steering arms. This process demonstrates advanced material manipulation and cost-reduction.



2. Tie Rods: Repurposed umbrella spokes were cut and bent to serve as rigid tie rods.

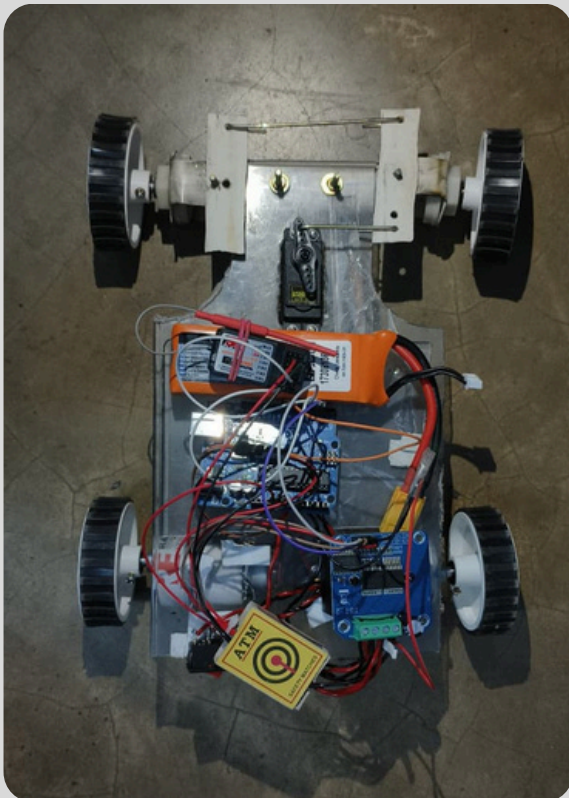
3. Bonding: High-strength adhesive (Fevikwik) was utilized in conjunction with mechanical fasteners to bond pivot points seamlessly.

This DIY approach highlights sustainable engineering, recycling, and practical problem-solving while maintaining mechanical integrity.

7) Estimated Weight

The use of a plastic chassis and a lithium-based battery pack keeps the overall weight well within standard competition limits:

- **Estimated Weight (without battery): 0.81kg**
- **Estimated Weight (with 3S battery pack): 1.00 kg**



(After assembling all components and making circuit connections)

8) Cost Report

The project was executed with strict budget constraints. All major and minor consumables are detailed below.

Components	Quantity	Unit Cost (Est.)	Total Cost
12V Johnson Geared DC Motors	2	380	760
BTS7960 43A Motor Driver	1	300	300
MG995R Metal Gear Servo	1	280	280
Arduino Uno	1	320	320
Flysky FS-CT6B Transmitter & Receiver	1	2600	2600
Duracell Battery (for Remote)	8	25	200
Robotic Rubber Wheels	4	30	120
3S1P 2200mAh Li-Po & Adapter	1 1	1650 + 300	1950
UBEC 5V / 3A	1	90	90
XT60	1	30	30
Consumables (Wires, Feviquick)	Assorted	-	70
Hardware (Nuts, Bolts, Switch)	Assorted	-	55
Total Estimated Cost			6775